

Predicting Employee Attrition with Machine Learning: Clustering, Classification, and Explainability on the IBM HR Analytics Dataset

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<https://github.com/dereksun64/IBM-HR-Attrition-Prediction>

Abstract

Employee attrition imposes substantial financial and operational costs on organizations, yet predicting which employees will leave remains a difficult classification problem due to severe class imbalance and heterogeneous workforce behavior. We present a complete machine-learning pipeline applied to the IBM HR Analytics dataset, comprising 1,470 employees described by 35 features and exhibiting a 16.12% attrition rate. Our pipeline spans exploratory data analysis, SMOTE-augmented preprocessing, K-Means unsupervised clustering for employee segmentation, and a systematic multi-round classification study covering Logistic Regression, Random Forest, and XGBoost under both oversampling and class-weight strategies, followed by hyperparameter tuning, stacking, and post-hoc SHAP explainability. K-Means clustering ($k=2$) reveals two latent segments with attrition rates of 19.2% and 9.5%, respectively. Among all classifiers, Logistic Regression trained with SMOTE achieves the highest F1 score of 0.5248 with recall of 0.7872, while XGBoost after hyperparameter tuning achieves the best ROC-AUC of 0.8132. SHAP analysis identifies `OverTime`, `YearsWithCurrManager`, and `EnvironmentSatisfaction` as the top attrition drivers by feature importance magnitude, providing actionable signals for targeted HR intervention.

1 Introduction

Employee turnover is among the costliest challenges facing modern organizations. Research consistently places the cost of replacing a single employee at 50–200% of their annual salary when recruiting costs and lost productivity are factored in [Fallucchi et al., 2020, Alduayj and Rajpoot, 2018]. Beyond direct cost, attrition erodes institutional knowledge and disrupts team dynamics. The ability to anticipate voluntary departures before they occur would allow HR departments to intervene with targeted retention programs, provided the predictive model is both accurate and interpretable.

The IBM HR Analytics dataset [IBM, 2015] provides a canonical benchmark for this problem: 1,470 employee records, 35 features spanning demographics, compensation, satisfaction ratings, and tenure, and a binary attrition outcome. With only 16.12% of employees having left (237 of 1,470), the dataset exemplifies the class imbalance that renders naive classification approaches useless in practice. A majority-class predictor achieves 84% accuracy while identifying zero leavers.

This paper makes three contributions:

1. **Unsupervised segmentation.** K-Means clustering partitions employees into two latent groups with markedly different attrition rates (19.2% vs. 9.5%), revealing structure not visible from aggregate statistics alone.
2. **Systematic classification benchmark.** We evaluate nine model configurations (spanning two imbalance-correction strategies, hyperparameter tuning, and a stacking ensemble), all benchmarked against a naive majority-class baseline, yielding clear guidance on which models and strategies best recover minority-class signal.
3. **SHAP-based explainability.** Post-hoc SHAP analysis on the best-performing model identifies the features most responsible for individual predictions. The top drivers by mean $|\text{SHAP}|$ magnitude are `OverTime`, `YearsWithCurrManager`, and `EnvironmentSatisfaction` (directional

effects verified by LR coefficients in Section 5). This enables data-driven HR intervention and diagnosis of systematic failure modes in the false-negative profile.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and preprocessing pipeline. Section 4 details the methodology. Section 5 presents experimental results, and Section 6 concludes.

2 Related Work

Employee and customer churn prediction. Machine learning has been applied extensively to churn prediction in both HR and customer analytics contexts. Alduayj and Rajpoot [2018] evaluated multiple classifiers on the IBM HR dataset and found ensemble methods superior to single models under standard accuracy metrics. Zhao et al. [2018] highlighted that class imbalance is the central challenge, comparing SMOTE, cost-sensitive learning, and decision-threshold tuning, and concluding that no single strategy dominates across all metrics. Fallucchi et al. [2020] benchmarked Logistic Regression, Decision Trees, and Random Forest on HR data and advocated for F1 score and ROC-AUC as the only reliable indicators when positive-class prevalence is below 20%. Our work extends this literature by conducting a rigorous multi-round comparison that isolates the independent effects of imbalance-correction strategy and hyperparameter tuning, enabling a cleaner attribution of performance differences.

Clustering for employee segmentation. Unsupervised methods have been used to identify latent employee groups prior to supervised modeling [Jain and Nayyar, 2018]. K-Means, despite its known sensitivity to initialization and difficulty with non-spherical clusters, remains the dominant baseline owing to computational efficiency and interpretability [MacQueen, 1967]. Low silhouette scores on HR data, a consistent finding in the literature, reflect the fact that employee features span qualitatively different scales and do not naturally form Euclidean clusters. We adopt K-Means for descriptive segmentation rather than as a classification feature, using PCA projection to visualize cluster structure and post-hoc attrition rates to interpret the groups.

Explainability with SHAP. Lundberg and Lee [2017] introduced SHAP (SHapley Additive ex-Planations), grounding feature attribution in cooperative game theory to produce consistent, locally accurate explanations. Unlike permutation importance, SHAP better handles correlated features than permutation importance by marginalizing over feature interactions. This is a critical property in HR datasets where tenure, income, and job level are heavily correlated (see Section 3). Recent HR analytics work has demonstrated that SHAP explanations not only improve model trust but can directly inform retention policy by identifying high-risk feature combinations at the individual level. We apply a LinearExplainer on our logistic regression model and examine both global feature importance rankings and per-instance waterfall plots to diagnose false-negative failures.

3 Dataset and Preprocessing

3.1 Dataset Description

The IBM HR Analytics dataset [IBM, 2015] contains records for 1,470 employees with 35 attributes per record. Features fall into five categories: *demographic* (Age, Gender, MaritalStatus, Distance-FromHome), *job-related* (JobRole, Department, JobLevel, BusinessTravel, OverTime), *compensation* (MonthlyIncome, DailyRate, HourlyRate, MonthlyRate, PercentSalaryHike, StockOptionLevel), *satisfaction* (JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, WorkLifeBalance, JobInvolvement), and *tenure* (YearsAtCompany, YearsInCurrentRole, TotalWorkingYears, YearsWith-CurrManager, YearsSinceLastPromotion, NumCompaniesWorked).

The binary target is *Attrition* (Yes/No). As shown in Figure 1, 237 employees left (16.12%) versus 1,233 who stayed (83.88%), constituting a class imbalance that makes accuracy an unreliable performance metric.

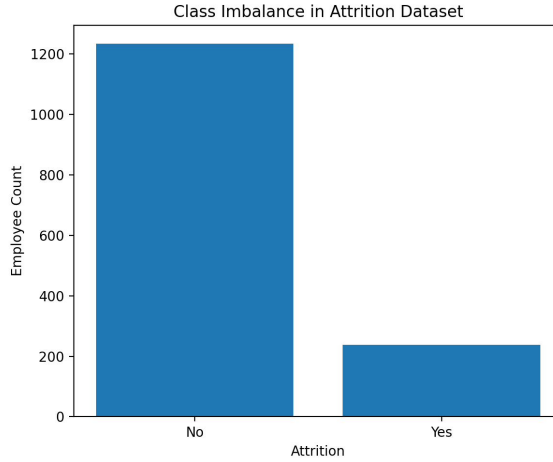


Figure 1: Class distribution of the Attrition target. Only 16.12% of employees left, creating a significant imbalance that motivates SMOTE-based oversampling and the use of F1 and ROC-AUC as primary evaluation metrics.

The dataset contains **no missing values**, as confirmed by our automated audit (Section 4). Four columns were dropped during preprocessing: three constant-valued columns (*EmployeeCount*, *StandardHours*, *Over18*), which carry no discriminative information, and *EmployeeNumber*, a unique employee identifier that was removed to prevent spurious memorization of index-based patterns (identifier leakage).

3.2 Preprocessing Pipeline

Encoding. Binary categorical features (e.g., *Gender*, *OverTime*) were label-encoded as 0/1. Multi-class categoricals (e.g., *Department*, *JobRole*, *MaritalStatus*, *EducationField*, *BusinessTravel*) were one-hot encoded.

Scaling. All numeric features were standardized using *StandardScaler* (zero mean, unit variance) after encoding. Scaling is a strict requirement for K-Means (distance-based) and Logistic Regression (gradient-based). Random Forest and XGBoost operate via threshold comparisons and are scale-invariant. Their split decisions and feature importance metrics are unaffected by linear rescaling, but received the same preprocessed input for pipeline consistency.

Train/test split. We applied a stratified 80/20 split with `random_state=42`, yielding 1,176 training and 294 test examples, preserving the 16.12% attrition rate in both partitions. The test set contains 247 employees who stayed and 47 who left.

SMOTE. Synthetic Minority Over-sampling Technique [Chawla et al., 2002] was applied exclusively to the training set after splitting. Applying SMOTE before splitting would introduce synthetic samples into the test set, allowing the model to “memorize” artificially generated minority examples and inflate evaluation metrics, a form of data leakage. By oversampling only the training fold, evaluation remains on the original, unaugmented distribution.

Correlation structure. Figure 2 shows the feature correlation heatmap. Seven pairs exceed the $|r| \geq 0.70$ threshold, with the strongest being *JobLevel*–*MonthlyIncome* ($r = 0.950$), *JobLevel*–*TotalWorkingYears* ($r = 0.782$), and *MonthlyIncome*–*TotalWorkingYears* ($r = 0.773$). Permutation importance distributes credit inconsistently among correlated features; SHAP handles this better, which is why we use it here.

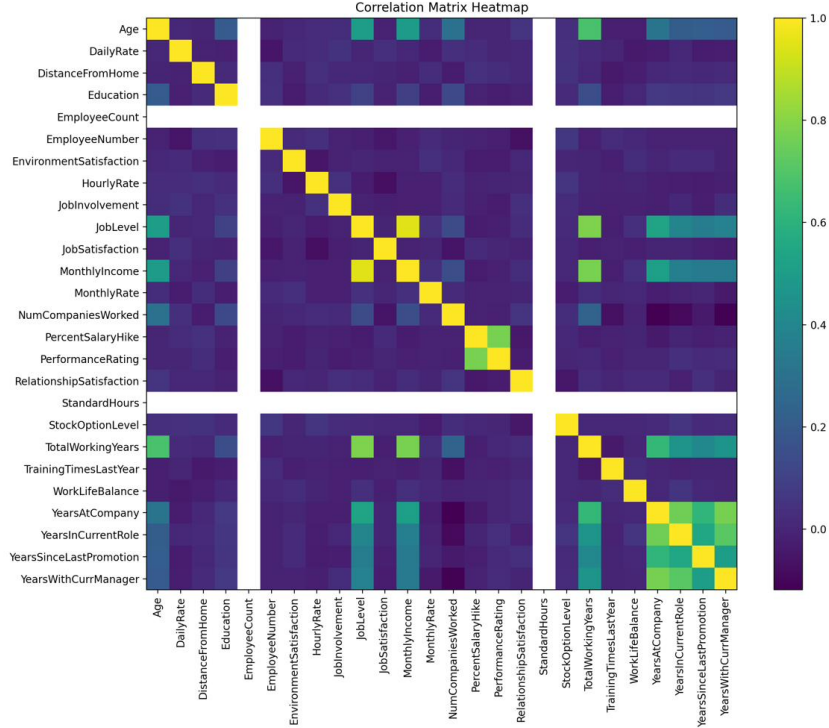


Figure 2: Feature correlation heatmap. Dark red cells indicate strong positive correlation. Seven pairs exceed $|r| = 0.70$, with JobLevel–MonthlyIncome showing the strongest dependence ($r = 0.950$).

4 Methodology

We evaluate nine model configurations across four experimental phases, summarized in Table 1. Each phase isolates one design variable; all models are evaluated on the same held-out test set.

Table 1: Overview of all nine model configurations.

Phase	Models	Imbalance strategy	Cluster feature?
Round 1	LR, RF, XGB	SMOTE (train only)	No
Round 2	LR, RF, XGB	Class weights	Yes (transductive)
Round 3 (tuning)	RF, XGB	Class weights + RandomizedSearchCV	No
Round 4 (stacking)	RF + XGB→LR	SMOTE (train only)	No

4.1 Naive Baseline

Before training any learned model, we established a majority-class baseline that always predicts *Attrition = No*. On the 294-example test set this achieves accuracy of 0.8401 while yielding $F1 = 0.0$ and $recall = 0.0$ (Table 2). This benchmark is essential: it demonstrates that accuracy is a misleading metric under class imbalance and motivates F1 score and ROC-AUC as our primary performance criteria. Any learned model that cannot surpass this baseline on F1 has failed to recover the minority-class signal that matters most for HR decision-making.

4.2 Unsupervised Clustering (K-Means)

K-Means [MacQueen, 1967] was applied to the full scaled feature matrix (excluding the Attrition label during fitting) to discover latent employee segments. The number of clusters k was chosen via two complementary criteria: the *elbow method*, which inspects the rate of inertia reduction as k increases, and the *silhouette score*, which measures how much better each point fits its assigned cluster than the nearest alternative.

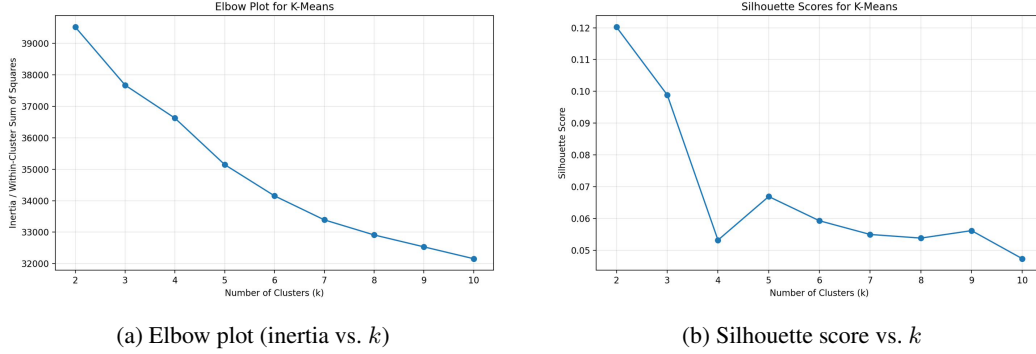


Figure 3: K-Means model selection over $k = 2, \dots, 10$. Inertia at $k = 2$ is 39,522; the silhouette score peaks at $k = 2$ (0.1202) and generally declines thereafter, with minor upticks at $k = 5$ (0.0669) and $k = 9$ (0.0562), confirming $k = 2$ as the overall optimal partition.

As shown in Figure 3, both criteria agree on $k = 2$, yielding inertia 39,522 and silhouette score 0.1202. In high-dimensional HR data, employee profiles rarely form globular, well-separated Euclidean clusters; features spanning demographics, compensation, and tenure create a feature space where clusters are naturally *soft and overlapping*. A low silhouette score therefore does not disqualify the result. What matters is whether the partition provides a meaningful lift in the outcome of interest. Here it does: Cluster 0 carries a 19.2% attrition rate versus 9.5% for Cluster 1, a two-fold differential that is both substantively large and operationally actionable even without individual-level prediction.

The cluster assignment was used *only* in Round 2 (appended as an extra feature alongside class-weight penalties) and was not included in Rounds 1, 3 (tuning), or 4 (stacking). Clustering therefore serves primarily as a descriptive tool for this report; the Round 2 results incorporating the cluster label should be compared against Round 1 with this design difference in mind.

4.3 Classification Pipeline

We trained three classifiers (Logistic Regression, RF, and XGBoost [Chen and Guestrin, 2016]) under four successive experimental rounds designed to isolate the effect of each design choice.

Round 1 (SMOTE + default hyperparameters). Models were trained on the SMOTE-balanced training set and evaluated on the original (unaugmented) test set. SMOTE increases minority-class representation during training without modifying the evaluation distribution.

Round 2 (class-weight correction + cluster feature). The same three models were retrained using class-weight penalties (`class_weight='balanced'` for LR and RF; `scale_pos_weight` for XGBoost), and the K-Means cluster label was additionally appended as a feature. Round 2 is best understood as an *exploratory composite strategy* rather than a controlled experiment: two design choices are changed simultaneously (imbalance correction method and the addition of a cluster feature), so individual contributions cannot be isolated from its results alone.

The cluster label introduces a mild form of transductive leakage because K-Means was fitted on the full 1,470-employee dataset, including test-set employees, before the train/test split. Concretely, the cluster assignment for a test employee was determined using that employee’s own feature values, information that would not be available to a real-time deployment system receiving unseen data. However, in this *offline analysis* context, the cluster label serves a legitimate purpose: it acts as a

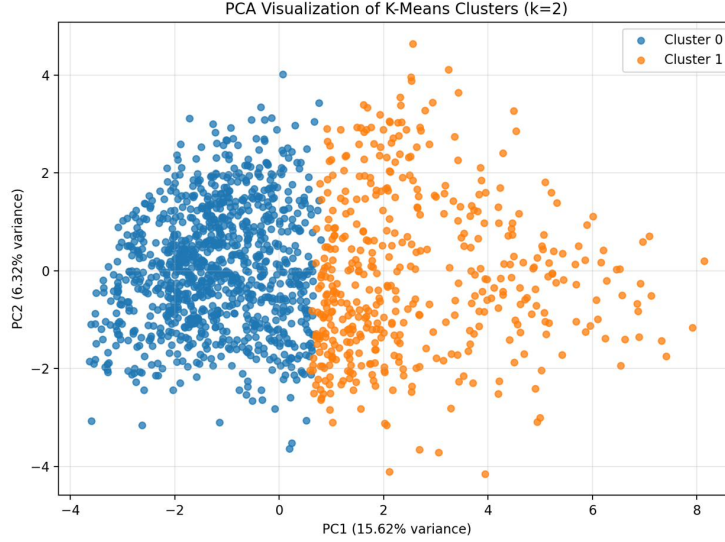


Figure 4: PCA projection of K-Means clusters ($k = 2$). Cluster 0 (higher attrition risk, 19.2%) and Cluster 1 (lower risk, 9.5%) overlap substantially in 2D PCA space, consistent with the low silhouette score and the high-dimensional, mixed-scale nature of the feature set.

proxy for latent workforce structure, compressing the full feature space into a single interpretable segment indicator. Round 2 metrics should therefore be read as an upper bound on what a properly deployed cluster feature could achieve, not as a reproducible out-of-sample estimate.

Round 3 (Hyperparameter tuning). The two ensemble models, RF and XGBoost, underwent `RandomizedSearchCV` with 5-fold cross-validation and 50 iterations. LR was excluded as a linear model with limited hyperparameter surface. Tuned models are evaluated on the held-out test set and compared against their Round 1 baselines throughout (Section 5) to provide a consistent reference point.

Stacking ensemble. A meta-learner (Logistic Regression) was trained on the out-of-fold predictions of two base learners: Random Forest with default hyperparameters and XGBoost with tuned hyperparameters (from Round 3). RF was left at default settings because tuning worsened its generalization (see above); using the tuned variant as a base learner would have embedded a worse model into the stack. The SMOTE-balanced training set was used throughout; the cluster feature was *not* included. Note that the meta-learner (Logistic Regression) is the same model family as LR Round 1, which is worth bearing in mind when interpreting any advantage the stack may provide over that baseline.

5 Experiments and Results

5.1 Main Results

Table 2 reports all nine model configurations alongside the naive baseline. Bold entries mark the best value in each column across learned models.

5.2 Classification Analysis

Logistic Regression (Round 1) achieves the best F1 and functions as a high-recall early-warning tool. With $F1 = 0.5248$ and $\text{recall} = 0.7872$, LR Round 1 correctly identifies 37 of the 47 true leavers in the test set (Figure 5), a detection rate of 78.7% compared to 0% for the naive baseline. For HR practitioners, this transforms an invisible problem into a manageable watchlist: the model

Table 2: Full results across all model configurations. Best values per column (excluding naive baseline) are **bolded**. Primary metrics are F1 and ROC-AUC, as accuracy is misleading under the 16.12% attrition rate.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Naive Baseline	0.8401	0.0000	0.0000	0.0000	0.5000
LR Round 1 (SMOTE)	0.7721	0.3936	0.7872	0.5248	0.8062
RF Round 1 (SMOTE)	0.8537	0.5909	0.2766	0.3768	0.7979
XGB Round 1 (SMOTE)	0.8639	0.6296	0.3617	0.4595	0.7928
LR Round 2 (weights)	0.7653	0.3778	0.7234	0.4964	0.8173
RF Round 2 (weights)	0.8435	0.5263	0.2128	0.3030	0.7806
XGB Round 2 (weights)	0.8503	0.5600	0.2979	0.3889	0.8068
RF Tuned	0.8469	0.5455	0.2553	0.3478	0.7836
XGB Tuned	0.8265	0.4643	0.5532	0.5049	0.8132
Stacking Ensemble	0.8333	0.4643	0.2766	0.3467	0.8030

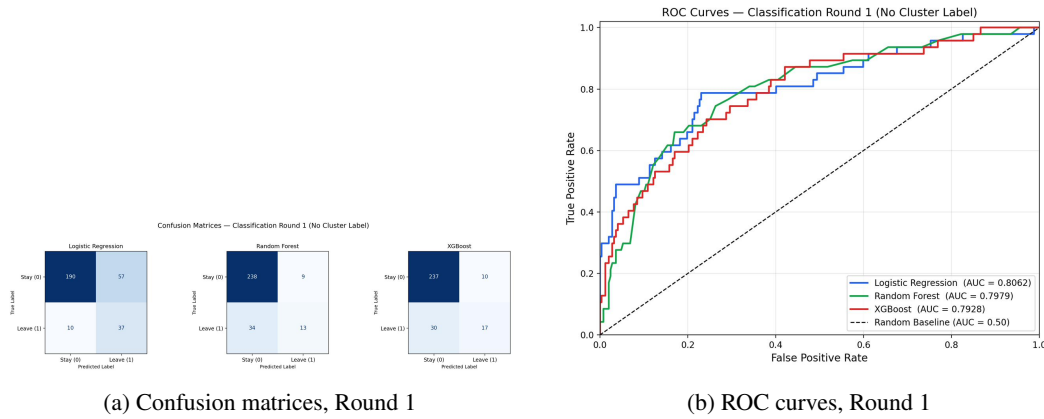


Figure 5: Round 1 results (SMOTE + default hyperparameters). Logistic Regression recovers the most true positives (37/47 leavers detected) at the cost of more false positives, yielding the best F1. RF and XGB exhibit higher precision but substantially lower recall.

surfaces nearly four in five employees who will leave, enabling targeted retention conversations before departures occur. The cost is increased false positives (57 employees incorrectly flagged), which depresses precision (0.394) and overall accuracy (0.772). In an HR context, however, a missed leaver is typically more costly than an unnecessary check-in. High recall is therefore the appropriate optimization target. SMOTE shifts the decision boundary toward the minority class by making the positive class appear more prevalent during training, producing the lower effective threshold that drives this recall improvement.

XGBoost (Tuned) achieves the best ROC-AUC. With ROC-AUC=0.8132, XGBoost’s tuned ensemble captures non-linear feature interactions that a linear boundary cannot represent. The ROC-AUC metric is threshold-agnostic: it measures the model’s ability to rank leavers above stayers across all thresholds. This makes XGBoost Tuned the preferred model when downstream applications require probability scores rather than hard predictions (e.g., for prioritizing targeted outreach).

F1 vs. ROC-AUC trade-off. These two metrics reflect different operational priorities. When HR teams need to flag a high-coverage list of at-risk employees (accepting some false alarms), LR Round 1 with its recall of 0.7872 is the appropriate choice. When the goal is to rank employees by departure probability for a constrained intervention budget, XGBoost Tuned’s ROC-AUC of 0.8132 provides the best ranking quality.

Round 2 (exploratory composite strategy) is competitive but not superior. LR Round 2

achieves a ROC-AUC of 0.8173 and an F1 of 0.4964. RF and XGB Round 2 underperform their Round 1 counterparts on F1. As noted in Section 4, Round 2 is an exploratory composite: two design choices change simultaneously, making it impossible to attribute performance differences to either factor alone. The transductive leakage further limits generalizability, so Round 2 metrics represent an upper bound on what a properly deployed cluster feature could achieve. The round is best read as a directional probe suggesting that adding segment structure does not meaningfully hurt performance, and may improve ranking quality at the margin.

Hyperparameter tuning improves XGBoost, not RF. Using Round 1 as the consistent baseline for both models: tuned XGBoost improves F1 from 0.4595 to 0.5049 (+0.045) and recall from 0.362 to 0.553, reflecting meaningful gains from expanded search over learning rate and tree depth. Tuned RF decreases F1 from 0.3768 to 0.3478 (−0.029), suggesting Random Forest reached a local optimum under default settings on this dataset and that further tuning did not help generalization.

Stacking does not outperform the best single model. The stacking ensemble (F1 = 0.347, ROC-AUC = 0.803) performs below both LR Round 1 and XGBoost Tuned. Because the two base learners (RF and tuned XGBoost) are both tree ensembles, they likely capture similar signals; the meta-learner Logistic Regression then has little diversity to exploit. Additionally, because LR (the meta-learner) is also the top-performing single model, any advantage of the stack over LR Round 1 would have to come from the base learners providing signal that LR cannot recover alone, which does not appear to be the case here.

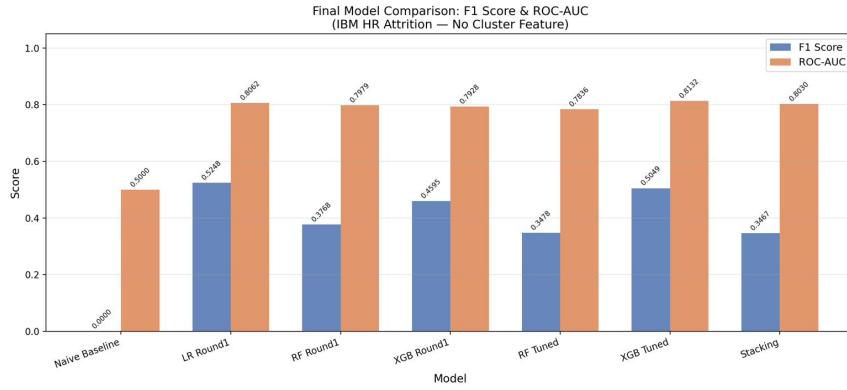


Figure 6: Comparison of model configurations on F1 and ROC-AUC, drawn from `final_results_summary.csv`. Round 2 models are excluded from this figure because they differ from other rounds along two confounded dimensions (imbalance strategy *and* cluster feature); they are included in Table 2 for completeness. LR Round 1 leads on F1; XGBoost Tuned leads on ROC-AUC.

5.3 Clustering Results

The K-Means partition reveals that Cluster 0 (68.4% of employees) carries a 19.2% attrition rate, nearly twice the 9.5% rate of Cluster 1 (31.6%). While the low silhouette score (0.1202) indicates that the clusters are not crisply separated in the feature space, the attrition rate differential is substantial enough to be actionable: HR resources directed at Cluster 0 members would intercept a disproportionate fraction of eventual leavers even without individual-level prediction. The PCA projection in Figure 4 confirms the overlap: the two clusters are not linearly separable in 2D, which motivates the subsequent supervised learning pipeline.

5.4 Error Analysis

The LR Round 1 model produces 10 false negatives (FN) and 57 false positives (FP) on the 294-example test set. **Model note:** the error analysis and SHAP scripts (`error_analysis.py`,

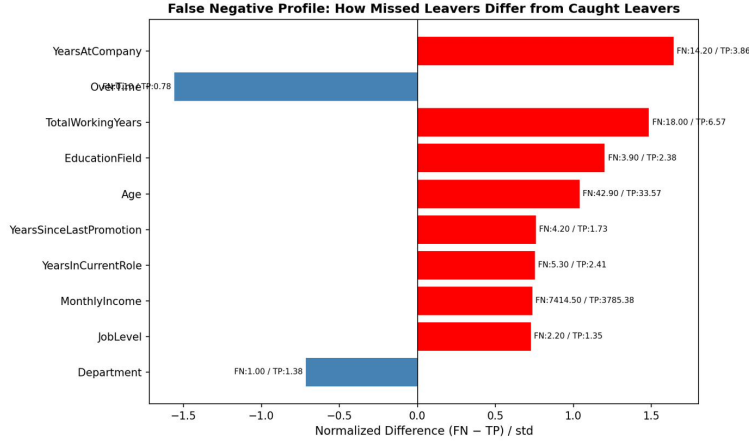


Figure 7: Feature profile of false negatives (employees who left but were predicted to stay). FN cases differ from true positives primarily along tenure and overtime dimensions.

explainability.py) retrain Logistic Regression with `class_weight='balanced'` on top of the already SMOTE-balanced training set, a reconstruction that differs from the Round 1 model in Table 2 (SMOTE only, no class weight). The resulting confusion matrix is identical (37 TP, 10 FN, 57 FP, 190 TN), so predicted labels agree, but SHAP coefficients belong to the reconstructed variant rather than the exact Round 1 model. Figure 7 profiles the FN subgroup against correctly identified leavers (true positives, TP). Three features show the largest normalized divergence: *YearsAtCompany* (FN mean: 14.20 vs. TP mean: 3.86), *OverTime* (FN mean: 0.10 vs. TP mean: 0.78), and *TotalWorkingYears* (FN mean: 18.00 vs. TP mean: 6.57).

Categorical FN rates show concentration in certain roles: 2 of 3 Manufacturing Directors (67%) and 1 of 1 Research Director (100%) in the test set were false negatives. Given the extremely small cell counts ($n = 3$ and $n = 1$ respectively), these should be treated as *case studies* rather than statistically significant trends, but they are consistent with the quantitative finding that the model systematically underperforms on senior employees.

Practical recommendation. The model is highly sensitive to burnout signals (particularly *OverTime*) and performs best on early-career employees whose attrition risk is visible in structured engagement metrics. HR teams should treat the model’s output as a first-pass triage for junior staff, while separately establishing manual monitoring programs for high-tenure directors and senior technical leads, whose attrition may stem from retirement planning, external poaching, or long-term disengagement not captured by the current feature set. The combination of model-flagged junior employees and manually monitored senior employees would provide broader coverage than either approach alone.

5.5 Explainability (SHAP)

SHAP LinearExplainer values were computed on the SMOTE-balanced training distribution [Lundberg and Lee, 2017], using `shap_values[1]` throughout. Positive SHAP values indicate a push toward leaving (class 1); negative values indicate a push toward staying.

Important caveat on directional inference. Using a SMOTE-augmented background means that for each feature, the SHAP baseline is the feature’s mean across synthetic-plus-real training samples rather than real employees alone. SMOTE interpolates new minority-class points between existing ones, which can shift the synthetic distribution’s feature means away from the real population mean. As a result, a feature may appear to have a positive mean SHAP value (implying it increases attrition risk) simply because the real test set has a *lower* mean for that feature than the inflated SMOTE baseline, rather than because the feature actually increases risk in the model. This artifact affected *YearsWithCurrManager* in our analysis, where the mean SHAP sign was positive (suggesting

increased risk) despite the LR coefficient being negative ($\beta = -0.536$, decreasing risk). Throughout this section, **directional influence is verified by the LR coefficients** (β), which are unaffected by the background distribution. Mean $|\text{SHAP}|$ values are used only to rank *magnitude* of influence.

Figure 8 and Appendix A reveal a clear hierarchy of feature influence. **OverTime** is by far the strongest predictor (mean $|\text{SHAP}| = 0.793$): working overtime substantially increases predicted attrition probability, consistent with literature on work–life balance burnout. **YearsWithCurrManager** (0.445) is the second-ranked feature by mean $|\text{SHAP}|$. Its LR coefficient is $\beta = -0.536$, indicating that longer tenure with the current manager *decreases* predicted attrition risk, consistent with the interpretation that an established manager relationship provides stability. The positive mean SHAP sign reported in the summary file reflects a SMOTE-background artifact (the synthetic training distribution has a higher mean for this feature than the real test set), and the coefficient is the correct directional reference. **EnvironmentSatisfaction** (0.438) decreases attrition risk when high ($\beta = -0.480$). **NumCompaniesWorked** (0.437) increases attrition risk when high ($\beta = +0.511$), consistent with the job-hopper pattern: employees with a history of frequent employer changes leave more readily. **TotalWorkingYears** (0.429) decreases risk when high ($\beta = -0.579$), indicating that more experienced employees exhibit lower attrition propensity on average.

Waterfall analysis of a representative false-negative case shows the competing SHAP forces that caused a misclassification. Under the positive-class sign convention, OverTime (-0.702) and Age (-0.653) pushed *toward staying*, while EnvironmentSatisfaction ($+0.721$) simultaneously pushed *toward leaving*. The net prediction was “stay” because the staying-signals collectively outweighed the leaving-signal, masking the true attrition outcome. In contrast, a correctly identified leaver received a strong net push toward leaving: OverTime ($+1.054$) and TotalWorkingYears ($+0.625$) drove the prediction upward while NumCompaniesWorked (-0.615) partially offset them, but the positive signals dominated ($+1.054 + 0.625 - 0.615 = +1.064$ net), yielding a confident correct prediction. This per-instance view underscores why global importance rankings are insufficient: competing SHAP effects can cancel at the instance level, and waterfall plots are the appropriate diagnostic for borderline or missed cases.

6 Conclusion

We presented a complete machine-learning pipeline for employee attrition prediction on the IBM HR Analytics dataset. Among nine model configurations benchmarked against a majority-class baseline ($F1 = 0$), Logistic Regression trained with SMOTE achieved the best minority-class recovery ($F1 = 0.5248$, $\text{recall} = 0.7872$), detecting 78.7% of leavers and serving as a practical high-recall early-warning tool for HR teams. XGBoost after hyperparameter tuning achieved the best probability ranking quality ($\text{ROC-AUC} = 0.8132$), making it the preferred model for constrained intervention budgets. K-Means clustering revealed two latent employee segments with attrition rates of 19.2% vs. 9.5%, providing an actionable segmentation even with a low silhouette score. SHAP analysis (with directional effects verified by LR coefficients to avoid SMOTE background artifacts) identified OverTime ($\beta = +0.796$, increases risk), YearsWithCurrManager ($\beta = -0.536$, decreases risk), and EnvironmentSatisfaction ($\beta = -0.480$, decreases risk) as the top three drivers by mean $|\text{SHAP}|$ magnitude. Error analysis revealed that the model systematically misses longer-tenured, low-overtime employees, motivating a hybrid monitoring strategy that combines model-flagged junior staff with manual oversight of senior roles. For future deployments, sweeping the classification threshold on a held-out validation set (rather than using the default 0.5) would be the single highest-leverage improvement available without retraining any model.

Limitations. Four limitations should be acknowledged. (1) *Weak clustering cohesion*: the silhouette score of 0.1202 is low, consistent with published HR clustering studies where mixed-scale tabular data rarely produces scores above 0.25 [Jain and Nayyar, 2018]. Density-based methods such as HDBSCAN may be better suited to this feature space. (2) *Single train/test split*: all metrics in Table 2 derive from one 80/20 split. With only 47 positive examples in the test set, a difference of one or two correct predictions shifts F1 by ≈ 0.02 – 0.04 ; repeated stratified k -fold cross-validation (e.g., 5×5) would be needed to report mean $\pm$ std and confirm that differences between models are not

split-specific noise. (3) *F1 ceiling at ≈ 0.52* : despite multiple tuning rounds, F1 did not exceed 0.53, suggesting the available features do not fully capture latent attrition causes. (4) *SMOTE-induced boundary distortion and background mismatch*: synthetic samples may not lie on the true data manifold, potentially distorting the decision boundary; additionally, using the SMOTE set as the SHAP background distribution (rather than the original training data) may shift feature attribution baselines. (5) *Fixed classification threshold*: all precision, recall, and F1 values are reported at the default 0.5 decision threshold. With SMOTE-shifted decision boundaries, this threshold is rarely optimal for imbalanced problems; a validation-set threshold sweep could meaningfully improve F1 and recall without retraining any model.

Future directions. (1) *Feature engineering*: interaction terms (e.g., OverTime $\times$ JobSatisfaction) and SHAP-guided feature selection could reduce noise and amplify discriminative signal. (2) *Alternative models*: CatBoost or a neural network with dropout regularization may better handle the feature correlations and non-linear interactions present in HR data. (3) *Longitudinal modeling*: a single cross-section captures only a snapshot; survival analysis or recurrent models trained on monthly employee snapshots would better capture the temporal dynamics of attrition risk and enable early intervention *before* disengagement becomes irreversible. (4) *Precision-recall curves and threshold tuning*: for an imbalanced problem with a 16% positive rate, precision-recall curves are more informative than ROC curves because they directly show the trade-off in the minority class. Adding PR curves for LR Round 1 and XGB Tuned, and sweeping the classification threshold on a validation set, would strengthen the experimental section and likely improve reported F1.

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A SHAP Feature Importance Bar Chart

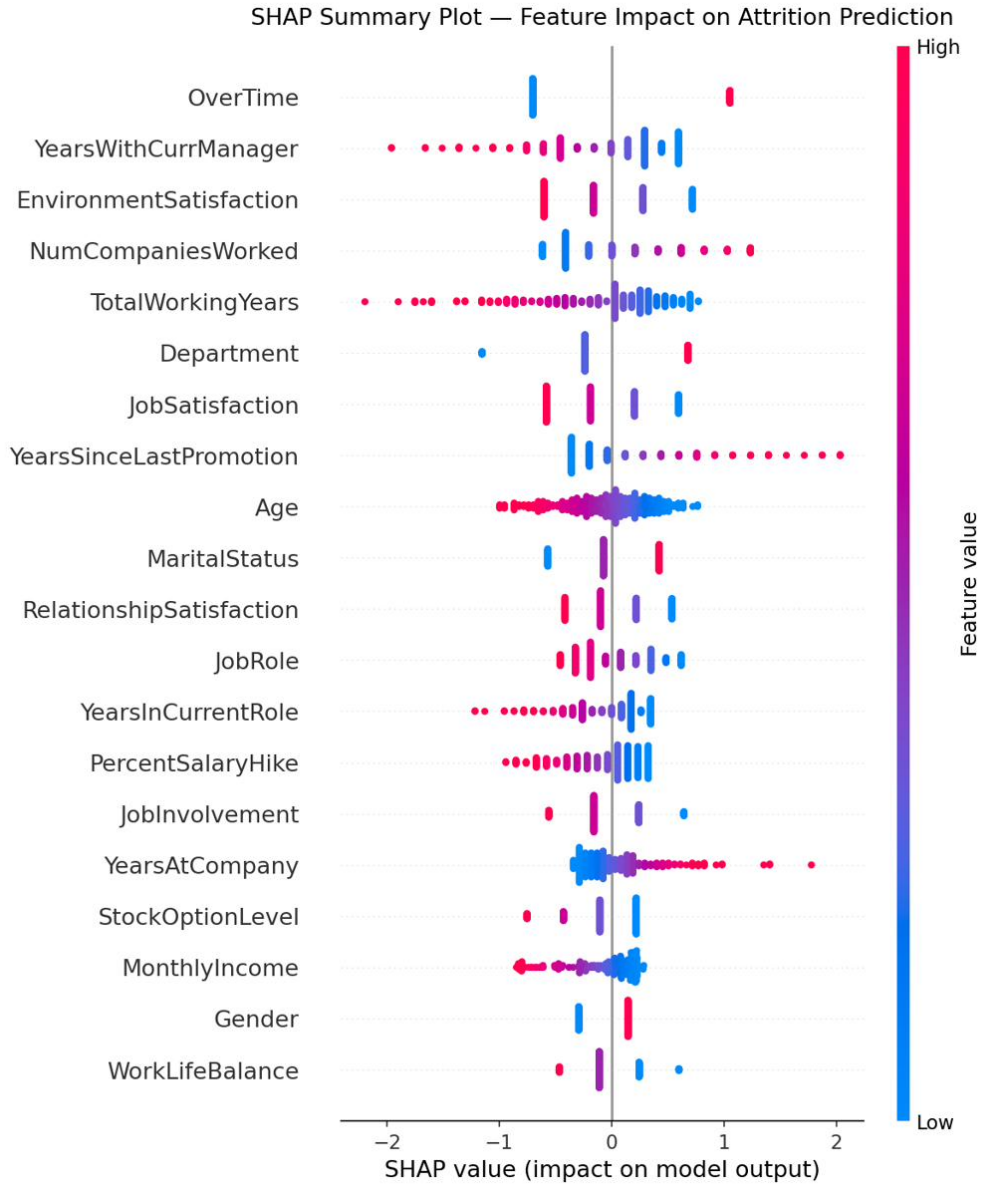


Figure 8: SHAP beeswarm plot for the reconstructed LR model. Each point represents one test employee; color encodes feature value (red = high, blue = low); horizontal position encodes the SHAP contribution toward leaving (positive = toward leave). OverTime is the dominant driver; the mean |SHAP| bar chart appears in Appendix A.

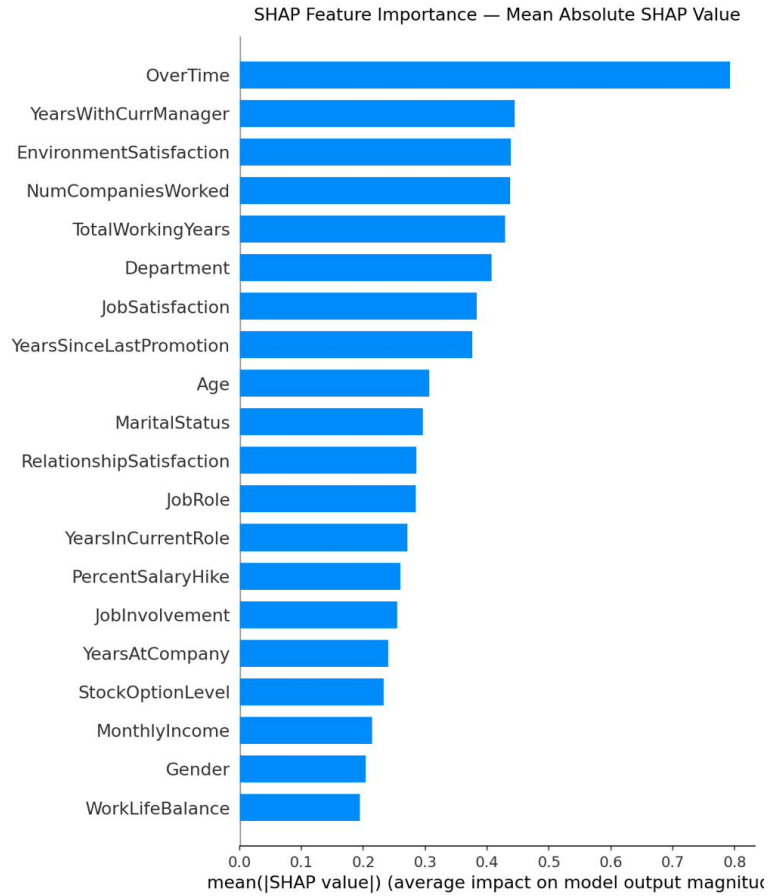


Figure 9: Global SHAP feature importance (mean $|\text{SHAP}|$) for the reconstructed LR model. Top five features by magnitude: OverTime (0.793), YearsWithCurrManager (0.445), EnvironmentSatisfaction (0.438), NumCompaniesWorked (0.437), and TotalWorkingYears (0.429). **Warning:** while mean $|\text{SHAP}|$ correctly indicates *importance magnitude*, the direction of influence should not be inferred from SHAP sign here. The SMOTE background distribution can invert the sign relative to the true model direction. Directional effects are verified by LR coefficients (β) as reported in Section 5.